# Lecture 7: Model Fitting

March 6, 2025

# Longitudinal Data Analysis

- **Scientific Questions:**
- **Data set**: $(Y_{ij}, ID_i, X_{1ij}, .., X_{pij})$ for $i = 1, \ldots, m$, and $j = 1, \cdots, n_i$
  [In the long format, we just add an extra column "ID" to the standard ("Y", "X1", .., "Xp") data format in statistics or machine learning]

- **Exploratory or Empirical Data Analysis**
  [wide or long format, plots, summary tables, etc.]

- **Confirmatory Data Analysis** (eg, ***statistical inference + prediction***)
  *** Model Fitting:**
  1. **Formulation**
  2. **Estimation**
  3. **Inference**
  4. **Diagnostics**
  5. **Model Selection**

# 1. Formulation of a Model

- Formulation of a model is a **continuation of exploratory or empirical data analysis**.
- First we need to address the **identification and/or outlier** issues.
  - Does the standard linear regression model $Y \sim X_1 + \cdots + X_p$ has collinearity or identification issues?
  - Are there any data with unusual large or small values?
- Second, we need to specify the model structure:
  - **Mean Structure (→ Level-1 Model or fixed effects)**
  - **Covariance Structure (→ Level-2 or random effects)**

# 1. Formulation of a Model (cont.)

- **Mean Structure (→ Level-1 model or fixed effects):**
  - **Time-plots** of observed averages for well replicated model;
  - Non-parametric local smoothing (e.g., **"loess"**) of the data when few measurements at any one time
  - Example of mean structure $Y_{ij} = h_{\theta_i}(\boldsymbol{X}_{ij}) + \epsilon_{ij}$ with $X_{ij}$ =Time/Wave
    - **Simple Linear model**: $\boldsymbol{h(x) = \beta_0 + \beta_1 x}$ with $\theta = (\beta_0, \beta_1)$
    - **Two-phase linear model**: $\boldsymbol{h(x) = u - \beta \max(0, v - x)}$ with $\theta = (\beta, v, u)$
    - **Broken stick model**: many piecewise linear models
    - **Logistic (S-shape Growth) model**: $\boldsymbol{h(x) = \phi_1 + \frac{\phi_2 - \phi_1}{1 + exp(-(x - \phi_3)/\phi_4)}}$ with $\theta = (\phi_1, \phi_2, \phi_3, \phi_4)$
    - **First-order open compartment model** for drug concentration: $\boldsymbol{h(x) = \frac{D\,\phi_1\phi_2}{\phi_3(\phi_2 - \phi_1)}\left[e^{-\phi_1 x} - e^{-\phi_2 x}\right]}$. Here $\phi_1$ =elimination rate from circulatory system, $\phi_2$ =absorption rate from digestive system, $\phi_3$ =clearance, $D$ =the dose.
    - **Multiple Linear regression model:** $\boldsymbol{Y_{ij} = \beta_{i0} + \beta_{i1}X_{1ij} + \cdots + \beta_{ip}X_{pij} + \epsilon_{ij}}$ (e.g., in survey data, $X_{1ij}$= Gender, $X_{2ij}$= Race,.., $X_{pij}$= interaction of (Exercise* Income), etc.) → variable selection!!!
    - Other linear or nonlinear models

# 1. Formulation of a Model (cont.)

- **Covariance Structure (→ Level-2 or random effects):**
  - Domain Knowledge
  - Which components of $\theta_i = (\theta_{i1}, \cdots, \theta_{ip})$ can be random effects?
    - We can start with the **intercept random effects** and then add more random effects. Often the intercept parameter is chosen to be random effects by default in the R program.
    - Or we can start to treat **all components as random effects with general correlation structure** to see whether the algorithm converges or not. If not, we can simplify the correlation structure by (ii) reducing the number of random effects or (ii) adding uncorrelation structure.
    - We can try multiple covariance models and then use AIC/BIC or other model selection criteria. → model selection!!!
  - How to model random-effects $\theta_i^* = (\theta_{i1}^*, \cdots, \theta_{iq}^*)$? It depends on how you model Level-1.
    - **For simple Level-1 model,** $\theta_i^* = \gamma_0 + \gamma_1 Z_{1i} + \cdots + \gamma_{1q} Z_{qi} + b_i$, **or** $\boldsymbol{\log(\theta_i^*) = \gamma_0 + \gamma_1 Z_{1i} + \cdots + \gamma_{1q} Z_{qi} + b_i}$
      **with** $\boldsymbol{b_i \sim MVN(0, \Sigma_b)}$ and $\boldsymbol{Z_{1i}, \cdots, Z_{qi}}$ **are some of those** $\boldsymbol{X_{ij} \equiv X_i}$
    - **Or for complicated level-1 models,** $\theta_i^*$ *or* $\log(\theta_i^*) \sim MVN(\theta, \Sigma_b)$.

# An Illustrative Example

**Data:** $(\boldsymbol{ID_i}, \boldsymbol{Y_{ij}}, \boldsymbol{t_{ij}}, \boldsymbol{X_i}), i = 1, \cdots, m, j = 1, \cdots, n_i$ [e.g., $\boldsymbol{X_i}$ = Gender]

**Models**: **Two equivalent ways** for the same composite model:

$$Y_{ij} = \beta_0 + \beta_1 t_{ij} + \beta_2 X_i + \beta_3 X_i t_{ij} + \boldsymbol{b_{0i}} + \boldsymbol{b_{1i} t_{ij}} + \epsilon_{ij}$$

- **Approach #1:**
  - Level-1 models: $\boldsymbol{Y_{ij} = \beta_{0i} + \beta_{1i} t_{ij} + \epsilon_{ij}}$
  - Level-2 models: $\boldsymbol{\beta_{0i} = \beta_0 + \beta_2 X_i + b_{0i}}$ and
    $\boldsymbol{\beta_{1i} = \beta_1 + \beta_3 X_i + b_{1i}}$
- **Approach #2:**
  - Level-1 models: $Y_{ij} = \beta_{0i} + \beta_{1i} t_{ij} + \beta_{2i} X_i + \beta_{3i} X_i t_{ij} + \epsilon_{ij}$
  - Level-2 models: $\beta_{0i} = \beta_0 + b_{0i},\ \beta_{1i} = \beta_1 + b_{1i}$, and
    $\beta_{2i} \equiv \beta_2, \beta_{3i} \equiv \beta_3$

# 2. Estimation

The second stage is model parameter estimation, i.e., attach numerical values to the parameters in the model.
**[Don't forget the variance parameters!]**

- Maximum Likelihood Estimator (MLE)
- Restricted Maximum Likelihood (REML) estimator

**Remarks:**

- **REML** reduces bias of variance estimate, thereby more **accurate statistical inference** of model parameters, although this is less important if the dim of $\beta$ is small w.r.t. the total sample size N
- **MLE** can produce **log-likelihood, deviance, AIC/BIC or other model selection criteria** for comparing different models, but the REML cannot!

# 3. Inference

- Recall that in the LME model, asymptotically
  $\hat{\beta} \sim MVN\left(\beta, \hat{V}\right)$ where $\hat{V} = \left[-\frac{\partial \log L(\beta)}{\partial\beta\partial\beta^T}\right]^{-1}$

- In the "lmer" package, given an "lme" object, say, "lme1", the R codes
  fixed(lme1) → $\hat{\beta} = (\hat{\beta}_1, \cdots, \hat{\beta}_p)$
  vcov(lme1) → $\hat{V}$ = p x p covariate matrix
  sqrt(diag(vcov(lme1))) → $(\hat{\sigma}_1, \cdots, \hat{\sigma}_p)$= standard errors

- How to conduct statistical inference on $\beta = (\beta_1, \cdots, \beta_p)$?

# 3(i) Hypothesis or CI on a given $\beta_j$

## Question: is the covariate $X_j$ associated with Y?

- For 95% confidence interval (CI) on a given $\beta_j$. With 95% probability

$$-1.96 \leq N(0,1) \leq 1.96$$

$$-1.96 \leq \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}_j} \leq 1.96$$

$$\hat{\beta}_j - 1.96\hat{\sigma}_j \leq \beta_j \leq \hat{\beta}_j + 1.96\hat{\sigma}_j$$

Thus, a 95% confidence interval for $\beta_j$ is $\left[\hat{\beta}_j - 1.96\hat{\sigma}_j,\ \hat{\beta}_j + 1.96\hat{\sigma}_j\right]$.
Is 0 value insides this CI or not??

- When we test the null hypothesis $H_0: \beta_j = 0$ against the alternative hypothesis $H_1: \beta_j \neq 0,$ the test statistic is

$$Z_{obs} = \frac{\hat{\beta}_j - 0}{\hat{\sigma}_j}$$

and at the significance level $\alpha = 5\%$,

  - reject $H_0: \beta_j = 0$ and declare $\beta_j \neq 0$ if and only if $|Z_{obs}| \geq 1.96$, or
  - claim we do not have enough evidence to reject $H_0: \beta_j = 0$ if $|Z_{obs}| < 1.96$

# 3(ii) CI on linear combination of $\beta$

- How to find the CI on $\psi = c_1\beta_1 + \cdots + c_p\beta_p$
  [e.g., $A + B$ or $3A - 2B$ in the ball weight problem]
- Key fact: $\hat{\psi} = c_1\hat{\beta}_1 + \cdots + c_p\hat{\beta}_p$

$$\hat{\psi} \sim N\left(\psi, \hat{\sigma}_{\psi}^2\right) \quad with \quad \hat{\sigma}_{\psi}^2 = \left(c_1, \cdots c_p\right)\hat{V}\begin{pmatrix} c_1 \\ \cdots \\ c_p \end{pmatrix}$$

- With 95% probability

$$-1.96 \leq N(0,1) = \frac{\hat{\psi} - \psi}{\hat{\sigma}_{\psi}} \leq 1.96$$

$$\psi \in \hat{\psi} \pm 1.96\, \hat{\sigma}_{\psi}$$

- R implementation via matrix operation by yourself:

```
cnew = as.matrix(c(c1,..,cp));
psi.hat = as.numeric( t(cnew) %*% as.marix( fixef(lme1) );
psi.sigma = as.numeric( sqrt(  t(cnew) %*% vcov(lme1) %*% cnew  )   );
psi.hat + c(-1,1) * 1.96* psi.sigma
```

# 3(iii) Confidence Region of $\beta$

- Suppose we want to find the joint confidence region of $r$ (or linear combinations) of $\beta$, say,

$$\psi_{r\times 1} = C_{r\times p}\beta_{p\times 1}$$

E.g., $C_{2\times p} = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \end{pmatrix}$ for $(\beta_1, \beta_2)$

- Fact:

$$\hat{\psi} \sim MVN\left( \psi, C\hat{V}C^T \right)$$
$$S(\psi) = \left(\hat{\psi} - \psi\right)^T \left[C\hat{V}C^T\right]^{-1} \left(\hat{\psi} - \psi\right) \sim \chi_r^2$$

- A $100(1-\alpha)\%$ confidence region for $\psi$ is

$$\{\psi : S(\psi) \leq c_r(q)\}$$

where $c_r(q)$ is the q-critical value of $\chi_r^2$.

- When testing $H_0 : \psi = 0$, it is sufficient to check whether 0 is inside this confidence region or not.

# 4. Diagnostics

- The aim is to compare the data with the fitted model in such a way to highlight any systematic discrepancies
- Since the LME models are essentially models for mean and covariance structures of data, simple and effective checks are to compare the fitted mean vs time-plot observed averages, etc.
- Other diagnosis plots from linear regression are also useful, e.g., QQ-normal plots (for level-1 residuals), and constant variances (for level-2 random-effects).

# 5. Model Selection

How to compare two different models?

- For **nested models,** conduct hypothesis testing
  - for fixed-effects, e.g, testing whether a fixed-effect coefficient is significantly different from 0 (e.g., via Z-statistics >= 1.96 from **REML or MLE**),
  - for random-effects, e.g., for the same fixed-effects, whether the variance of some random effects are zero or not (**via ANOVA from MLE**!)

- For two models with different fixed-effects and different random–effects, in general, use the values of the **loglikelihood, deviance, and AIC/BIC** criterion from **MLE.** (the smaller AIC/BIC, the better)

# More Example

- Let us go through more examples

- Below are Dr. Cook’s slides

# This Lecture

- Step 1: Fitting the unconditional models
  - Unconditional mean model
  - Unconditional growth model
- Step 2: Model building to the “final model”
  - Predictor only model
  - Predictor with control variable model
  - “Final” model

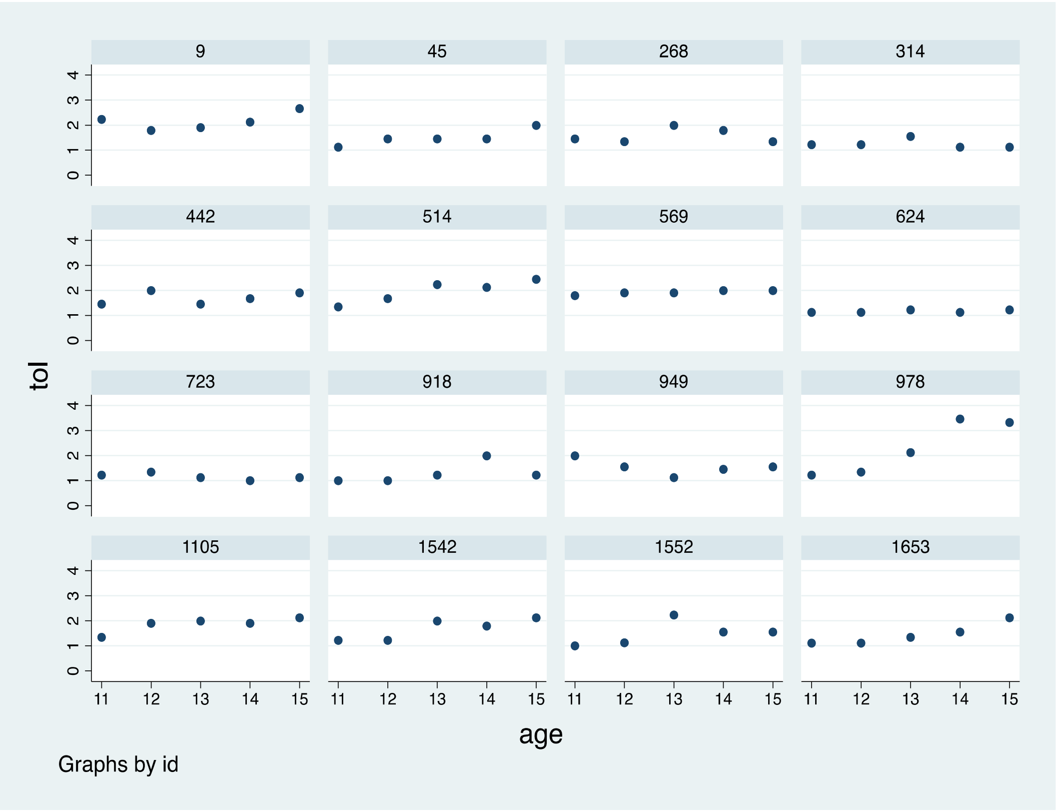
# Person-Period ("long") data format

(Singer and Willet page 25)

- Descriptive analyses
  - How do people change over time?
  - Is there growth?
  - What is the functional form?

- Empirical growth plot
  - Who is increasing, who is decreasing, who is increasing the most?
  - What information can you garner from this plot?

```
. graph twoway scatter tol age, by(id) ylabel(0(1)4) xlabel(11(1)15)
```



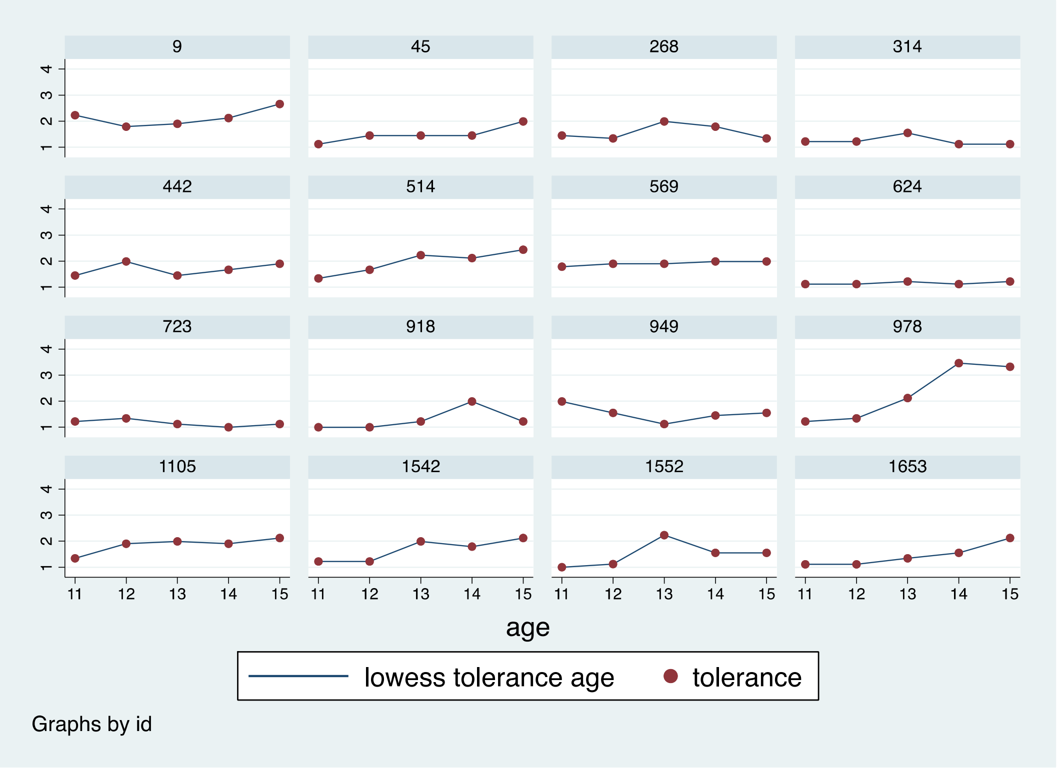
# Person-Period ("long") data format

(Singer and Willet pages 26 and 27)

- Non-parametric standardized approach
  - No functional form
  - "Let the data speak for themselves"

- What do you notice about the data?
  - Is the trajectory a bit clearer now?

. graph twoway (lowess tolerance age)(scatter tolerance age), by(id)

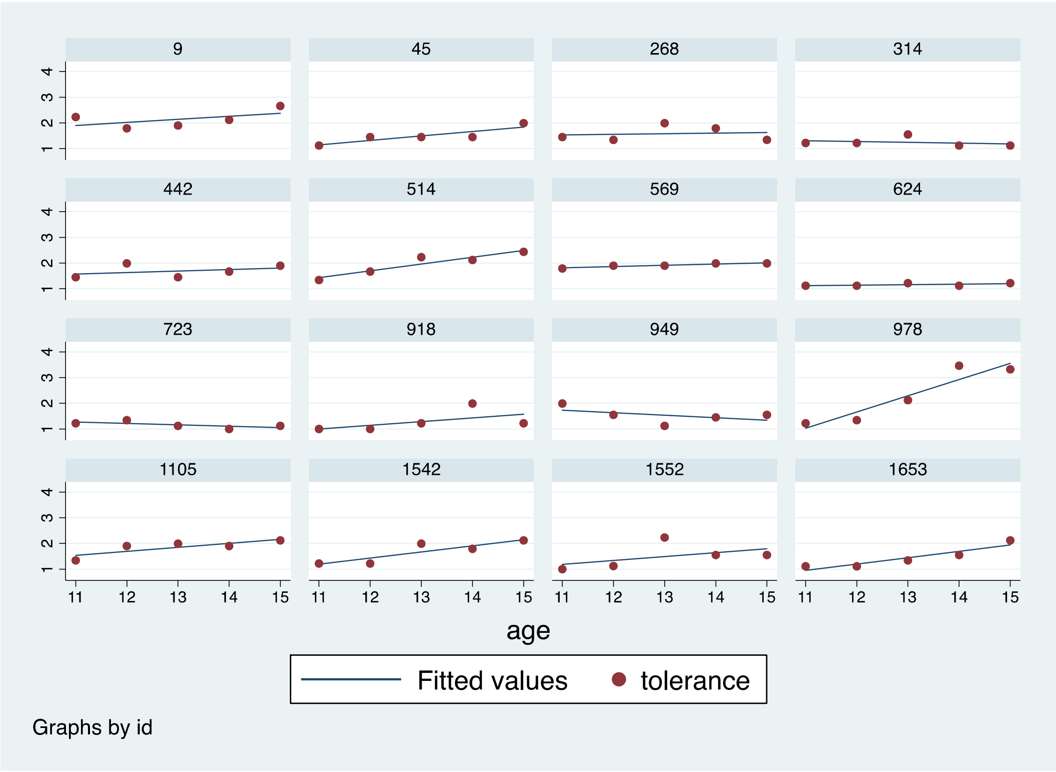


# Person-Period ("long") data format

(Singer and Willet pages 32-34)

- Parametric standardized approach
  - OLS linear regression
  - Fitted OLS trajectories superimposed on empirical growth plots for participants
- What do you notice about the data?
  - Anything different?
  - Confirmatory?

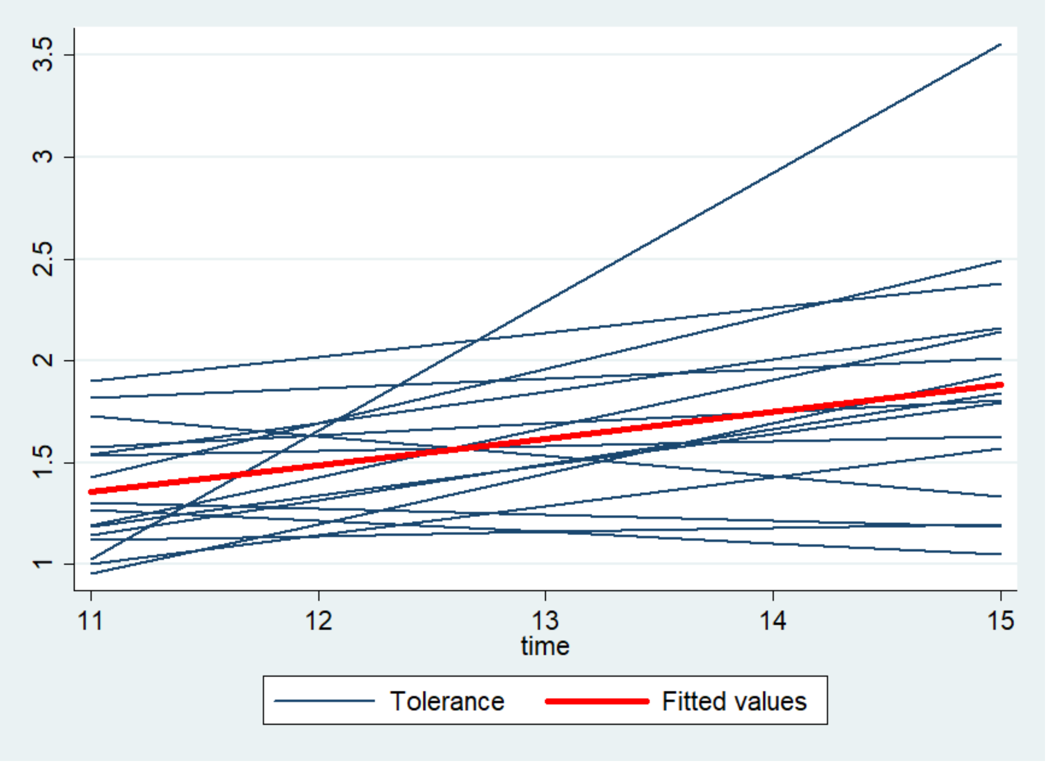
. graph twoway (lfit tolerance age)(scatter tolerance age), by(id)



# Person-Period ("long") data format

(see dofile in Lecture 2 folder, Singer and Willet pages 34 and 38)

- Parametric standardized approach
  - OLS linear regression
  - Fitted OLS trajectories superimposed on empirical growth plots for participants
  - **Overall mean trajectory line**
- What do you notice about the data?
  - Anything different?
  - Confirmatory?



# Unconditional mean model

$$Y_{ij} = \pi_{0i} + \varepsilon_{ij}$$
$$\pi_{0i} = \gamma_{00} + \zeta_{0i}$$

OR

$$Y_{ij} = \gamma_{00} + \zeta_{0i} + \epsilon_{ij}$$

- $\gamma_{00}$ = grand mean (fixed effect)
- $\zeta_{0i}$ = the amount person i's mean deviates from the population mean (between-person)
- $\epsilon_{ij}$ = the amount the response on occasion j deviates from person i's mean (within-person)
- $\epsilon_{ij} \sim N(0, \sigma_\epsilon^2)$
- $\zeta_{0i} \sim N(0, \sigma_0^2)$

$\zeta_{0i}$ = the distance between person i's mean and and the population average mean

- The unconditional means represent the <u>observed</u> value of **Y** for individual **i** on occasion **j, which** is composed of deviations about the <u>person-specific</u> and the grand means

# Unconditional growth model

$$Y_{ij} = \pi_{0i} + \pi_{1i} TIME_{ij} + \varepsilon_{ij}$$
$$\pi_{0i} = \gamma_{00} + \zeta_{0i}$$
$$\pi_{1i} = \gamma_{10} + \zeta_{1i},$$

$$Y_{ij} = \gamma_{00} + \gamma_{10} * time_{ij} + \zeta_{0i} + \zeta_{1i} * time_{ij} + \epsilon_{ij}$$

- $\gamma_{00}$ = average intercept (fixed effect)
- $\gamma_{10}$ = average slope (fixed effect)
- $\zeta_{0i}$ = the amount person i's intercept deviates from the population intercept
- $\zeta_{1i}$ = the amount person i's slope deviates from the population slope
- $\epsilon_{ij}$ = the amount the response on occasion j deviates from person i's true change trajectory
- $\epsilon_{ij} \sim N(0, \sigma_\epsilon^2)$ — $\sigma_\varepsilon^2$ Summarizes the scatter of each persons data around his or her own linear change trajectory
- $\zeta_{0i} \sim N(0, \sigma_0^2)$ — $\sigma_0^2$ Summarizes the between person variability
- $\zeta_{1i} \sim N(0, \sigma_1^2)$ — $\sigma_1^2$ Summarizes the within person variability
- $\zeta_{0i}$ and $\zeta_{1i}$ have covariance $\rho\sigma_1\sigma_2$

# Learning Objectives

- Clarifications
  - Unconditional mean model
  - Descriptive Statistics
- Assessing model fit
  - AIC/BIC
  - Loglikelihood
  - Deviance
- Testing model assumptions
  - Checking functional form
  - Checking normality
  - Checking homoscedasticity
  - Testing model assumptions
- Group project review

# Assessing Model Fit

# Assessing Model Fit

- Loglikelihood
  - Just like in regression, we want to see the overall value of the LL increase (if negative get closer to 0) as we move from our mean only model to our "final" model
- Deviance statistic
  - Difference between the current model and saturated model (exactly the same as the residual sum of squares in linear regression model)
- AIC/BIC
  - Just like in regression, we examine the absolute value of both. When running several nested models, we would like to see a decrease in the AIC/BIC

# Loglikelihood

- When the absolute value of the -2LL decreases this is an indication of a better model fit
- Lets take a look at the Cook et al paper from last weeks class exercise
  - Which is the “best model” according to the data?
- The loglikelihood will always increase (if negative, decrease or get closer to zero) when additional variables are added
  - This is why we examine multiple fit indicators in tandem

# Deviance

(Singer and Willett, pgs. 116-120)

(4.15)

$$\text{Deviance} = -2[LL_{\text{current model}} - LL_{\text{saturated model}}].$$

$$\text{Deviance} = -2LL_{\text{current model}}.$$

- Deviance statistic compares the loglikelihood of the current and saturated model
- How much worse is the current model than the best model (the saturated model)
- A small deviance means the current model is a close fit with the saturated model. This indicates a nice model fit
- A large deviance mean the current model is a poor fit

# Deviance

- To compare models….
  - One must be nested in the other
  - The same data must be used
  - The same number of cases must be used in both models
  - If FML comparing entire models
  - If REML only comparing the random effects

# Deviance Example

- Mean model only
- -2LL = -335.07799

Per the Deviance formula we multiple by 2, which = **670.16**

```
Mixed-effects ML regression                     Number of obs      =       246
Group variable: id                              Number of groups   =        82

                                                Obs per group:
                                                           min =         3
                                                           avg =       3.0
                                                           max =         3

                                                Wald chi2(0)       =         .
Log likelihood = -335.07799                     Prob > chi2        =         .

        alcuse |      Coef.    S
---------------+------------------
         _cons |   .9219549    .

  Random-effects Parameters
-------------------------------
id: Identity
                   var(_cons)
-------------------------------
                var(Residual)
-------------------------------
LR test vs. linear model: chi
```

- Growth model
- -2LL = -318.83122
- Per the Deviance formula we multiple by 2, which = **636.61**

```
Mixed-effects ML regression                     Number of obs      =       246
Group variable: id                              Number of groups   =        82

                                                Obs per group:
                                                           min =         3
                                                           avg =       3.0
                                                           max =         3

                                                Wald chi2(1)       =     20.11
Log likelihood = -318.83122                     Prob > chi2        =    0.0000

------------------------------------------------------------------------------
      alcuse |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
      age_14 |   .2706514    .060358     4.48   0.000     .1523519    .3889509
       _cons |   .6513035   .1015525     6.41   0.000     .4522642    .8503428
------------------------------------------------------------------------------

------------------------------------------------------------------------------
  Random-effects Parameters  |   Estimate   Std. Err.     [95% Conf. Interval]
-----------------------------+------------------------------------------------
id: Independent              |
                 var(age_14) |   .1172599   .0446745      .0555715    .2474266
                  var(_cons) |   .5432039   .1170779      .3560423    .8287513
-----------------------------+------------------------------------------------
               var(Residual) |   .3629466   .0533485       .272099    .4841261
------------------------------------------------------------------------------
LR test vs. linear model: chi2(2) = 78.65                 Prob > chi2 = 0.0000
```

# Deviance Example

**670.16 - 636.61 = 33.5**

**33.5 > 16.27** (the .001 critical value for the $X^2$ distribution)

**We conclude that the unconditional growth model provides a better fit than the mean only model**

```
Mixed-effects ML regression                     Number of obs      =       246
Group variable: id                              Number of groups   =        82

                                                Obs per group:
                                                              min =          3
                                                              avg =        3.0
                                                              max =          3

                                                Wald chi2(0)       =          .
Log likelihood = -335.07799                     Prob > chi2        =          .
```

```
Mixed-effects ML regression                     Number of obs      =       246
Group variable: id                              Number of groups   =        82

                                                Obs per group:
                                                              min =          3
                                                              avg =        3.0
                                                              max =          3

                                                Wald chi2(1)       =     20.11
Log likelihood = -318.83122                     Prob > chi2        =    0.0000
```

| alcuse | Coef. | Std. Err. | z | P>\|z\| | [95% Conf. | Interval] |
|---|---|---|---|---|---|---|
| age_14 | .2706514 | .060358 | 4.48 | 0.000 | .1523519 | .3889509 |
| _cons | .6513035 | .1015525 | 6.41 | 0.000 | .4522642 | .8503428 |

| Random-effects Parameters | Estimate | Std. Err. | [95% Conf. | Interval] |
|---|---|---|---|---|
| id: Independent | | | | |
| var(age_14) | .1172599 | .0446745 | .0555715 | .2474266 |
| var(_cons) | .5432039 | .1170779 | .3560423 | .8287513 |
| var(Residual) | .3629466 | .0533485 | .272099 | .4841261 |

LR test vs. linear model: chi2(2) = 78.65 Prob > chi2 = 0.0000

# AIC/BIC

(Singer and Willett, pgs. 120-122)

- Akaike Information Criterion (AIC)/ Bayesian Information Criterion (BIC)
  - Correct for some of the limitations of singularly using the loglikelihood and the Deviance statistic
- AIC controls for the increase in model parameters
- BIC controls for the number of parameters and sample size
- These models do not need to be nested as long as the same data is utilized
- The smaller the AIC/BIC the better the model

# AIC/BIC

(Singer and Willett, pgs. 120-122)

- Under MLE, can use both fixed and random effect parameters
- Under REML only random effects
- Art and science
  - Not clear what is a significant drop
  - Should always be used together
  - Conceptual model important

# Notes on Model Fit

- In Stata you can use the code **estat ic** to get some of the model fit statistics we just discussed
- Model fit statistics should be taken together. When all are trending in a specific direction, then you have more confidence in your model fit
- There are many more model fit techniques that we will discuss throughout the rest of the course, but these are the basics

# *A final comment about estimation and hypothesis testing*

***Two most common methods of estimation***

***Maximum likelihood (ML):***
*Seeks those parameter estimates that maximize the likelihood function, which assess the joint probability of simultaneously observing all the sample data actually obtained (implemented, e.g., in HLM and SAS Proc Mixed).*

***Generalized Least Squares (GLS): (& Iterative GLS)****: Iteratively seeks those parameter estimates that minimize the sum of squared residuals (allowing them to be autocorrelated and heteroscedastic) (implemented, e.g., in MLwiN and stata xtreg).*

***A more important distinction: Full vs. Restricted (ML or GLS)***

***Full:*** ***Simultaneously estimate the fixed effects and the variance components.***

- *Default in MLwiN & HLM*

***Goodness of fit statistics apply to the entire model***
***(both fixed and random effects)***
***This is the method we've used in both the examples shown so far***

***Restricted:*** ***Sequentially estimate the fixed effects and then the variance components***

- *Default in SAS Proc Mixed & stata xtmixed*

***Goodness of fit statistics apply to only the random effects***
***So we can only test hypotheses about VCs (and the models being compared must have identical fixed effects)***

# Testing Model Assumptions

# Testing Model Assumptions

(Singer and Willett, pgs. 127-137; Table 4.3; pg. 129)

- Checking functional form
  - Is the form linear, binomial, cubic, etc.
- Checking normality
  - Normality of the residual is key to many of the assumptions used for the random effects and for MLE
- Checking homoscedasticity
  - Key assumption of the random effects

**Table 4.3: Strategies for checking assumptions in the multilevel model for change, illustrated using Model F of tables 4.1 and 4.2 for the alcohol use data**

| Assumption and what to expect is tenable | What we find in the alcohol use data | | |
|---|---|---|---|
| | level-1 residual, $\varepsilon_{ij}$ | level-2 residual, $\zeta_{0i}$ | level-2 residual, $\zeta_{1i}$ |
| *Shape*. Linear individual change trajectories and linear relationships between individual growth parameters and level-2 predictors. | Empirical growth plots suggest that most adolescents experience linear change with age. For others, the small number of waves of data (3) makes it difficult to declare curvilinearity making the linear trajectory a reasonable approximation. | Because *COA* is dichotomous, there is no linearity assumption for $\pi_{0i}$. With the exception of two extreme data points, the plot of $\pi_{0i}$ vs. *PEER* suggests a strong linear relationship. | Because *COA* is dichotomous, there is no linearity assumption for $\pi_{1i}$. Plot of $\pi_{1i}$ vs. *PEER* suggests a weak linear relationship. |
| *Normality*. All residuals, at both level-1 and level-2, will be normally distributed. | A plot of $\varepsilon_{ij}$ vs. normal scores suggests normality. We find further support for normality in a plot of standardized $\varepsilon_{ij}$ vs *ID*, which reveals no unusual data points. | A plot of $\zeta_{0i}$ vs. normal scores suggests normality. So does a plot of standardized $\zeta_{0i}$ vs. *ID*, which reveals no unusual data points. There is slight evidence of a floor effect in the outcome. | A plot of $\zeta_{1i}$ vs. normal scores suggests normality, at least in the upper tail. The lower tail seems compressed. We find further support for this claim when we find no unusual data points in a plot of standardized $\zeta_{1i}$ vs. *ID*. There is also evidence of a floor effect in the outcome. |
| *Homoscedasticity*. Equal variances of the level-1 and level-2 residuals at each level of every predictor. | A plot of $\varepsilon_{ij}$ vs. AGE suggests approximately equal variability at ages 14, 15, and 16. | A plot of $\zeta_{0i}$ vs. *COA* suggests homoscedasticity at both values of *COA*. So does a plot vs. *PEER*, at least for values up to, and including, 2. Beyond this, there are too few cases to judge. | A plot of $\zeta_{1i}$ vs. *COA* suggests homoscedasticity at both values of *COA*. So does a plot vs. *PEER* at least for values up to, and including, 2. Beyond this, there are too few cases to judge. |

# Checking Functional Form

(Singer and Willett, pg. 128)

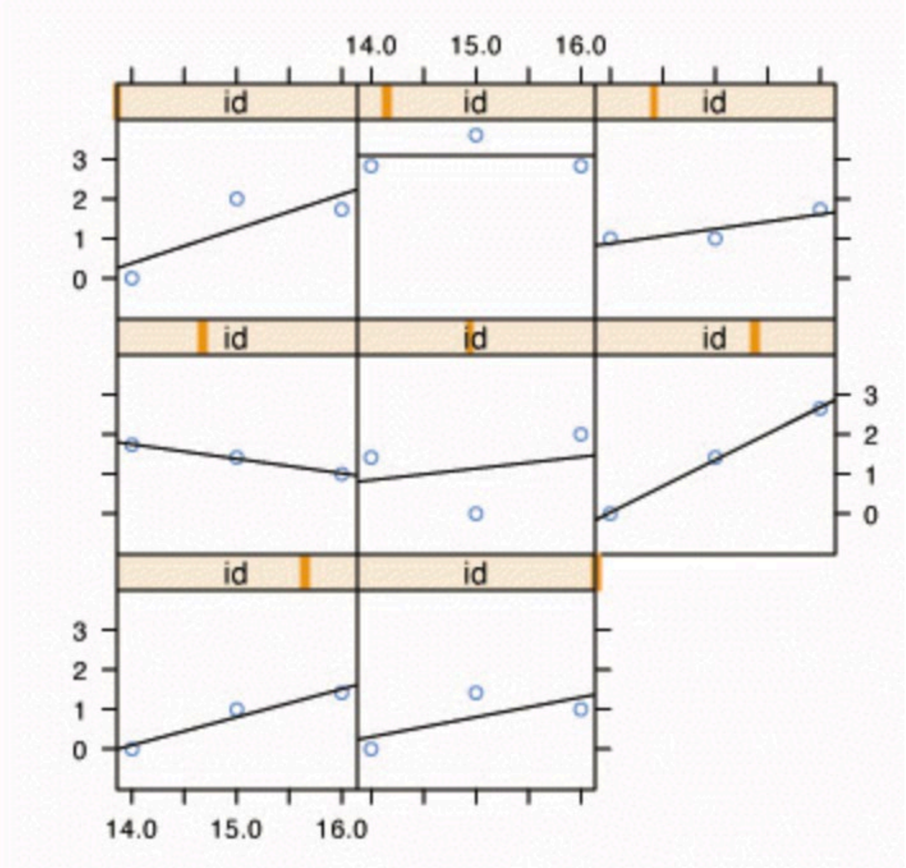
- **At level-1.** For each individual, examine empirical growth plots and superimpose an OLS-estimated individual change trajectory. Inspection should confirm the suitability of its hypothesized shape.
- **At level-2.** Plot OLS estimates of the individual growth parameters against each level-2 predictor. Inspection should confirm the suitability of the hypothesized level-2 relationships.

# Checking Functional Form
## (Singer and Willett, pg. 128)

**# 1**

- Overall individual growth plots look linear to me.
  - You can take another random sample and see if you get the same overall pattern

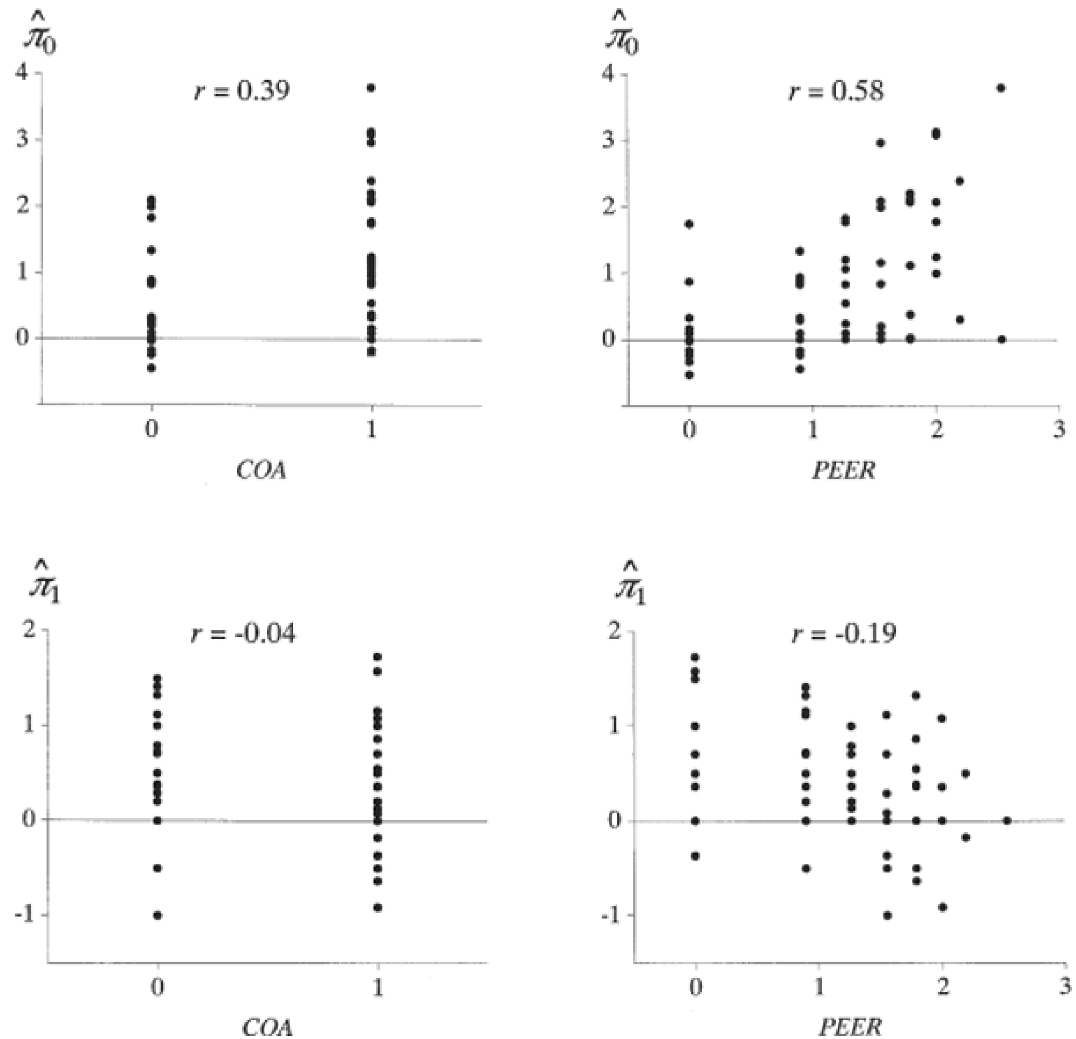
```
library(lattice)

xyplot(alcuse~age | id,
   data=alcohol1[alcohol1$id %in% c(4, 14, 23, 32, 41, 56, 65, 82), ],
   panel=function(x,y){
        panel.xyplot(x, y)
        panel.lmline(x,y)
}, ylim=c(-1, 4), as.table=T)
```



# Checking Functional Form
## (Singer and Willett, pg. 128)

**#2**

- Overall the OLS estimates look linear for each of the 2 time-invariant covariates
- When covariate dichotomous, nothing really to assess. You can see if there are any outliers
- The two plots for peers seem linear with some outliers (scatter about with no clear form)



# Checking Functional Form

(Singer and Willett, pg. 128)

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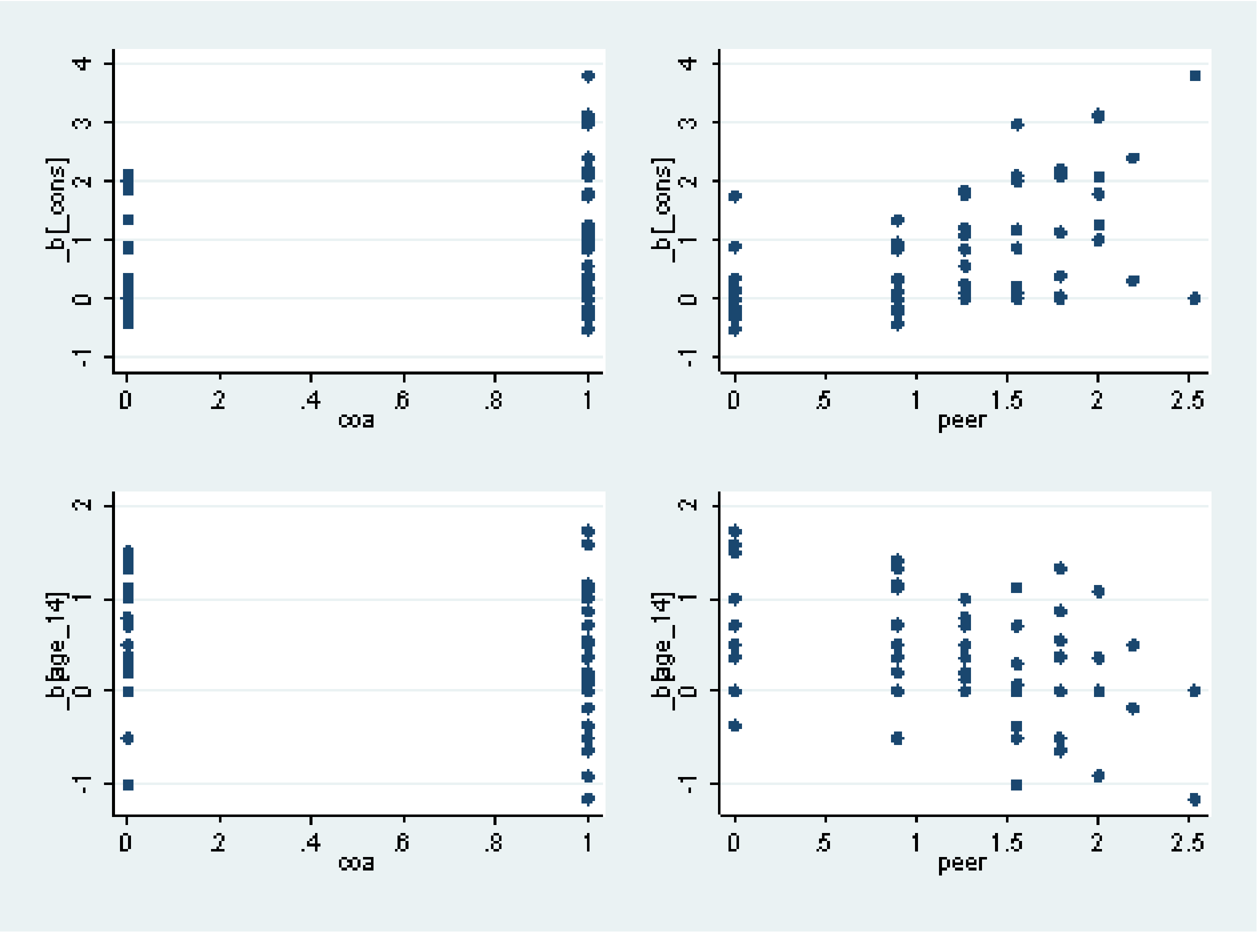
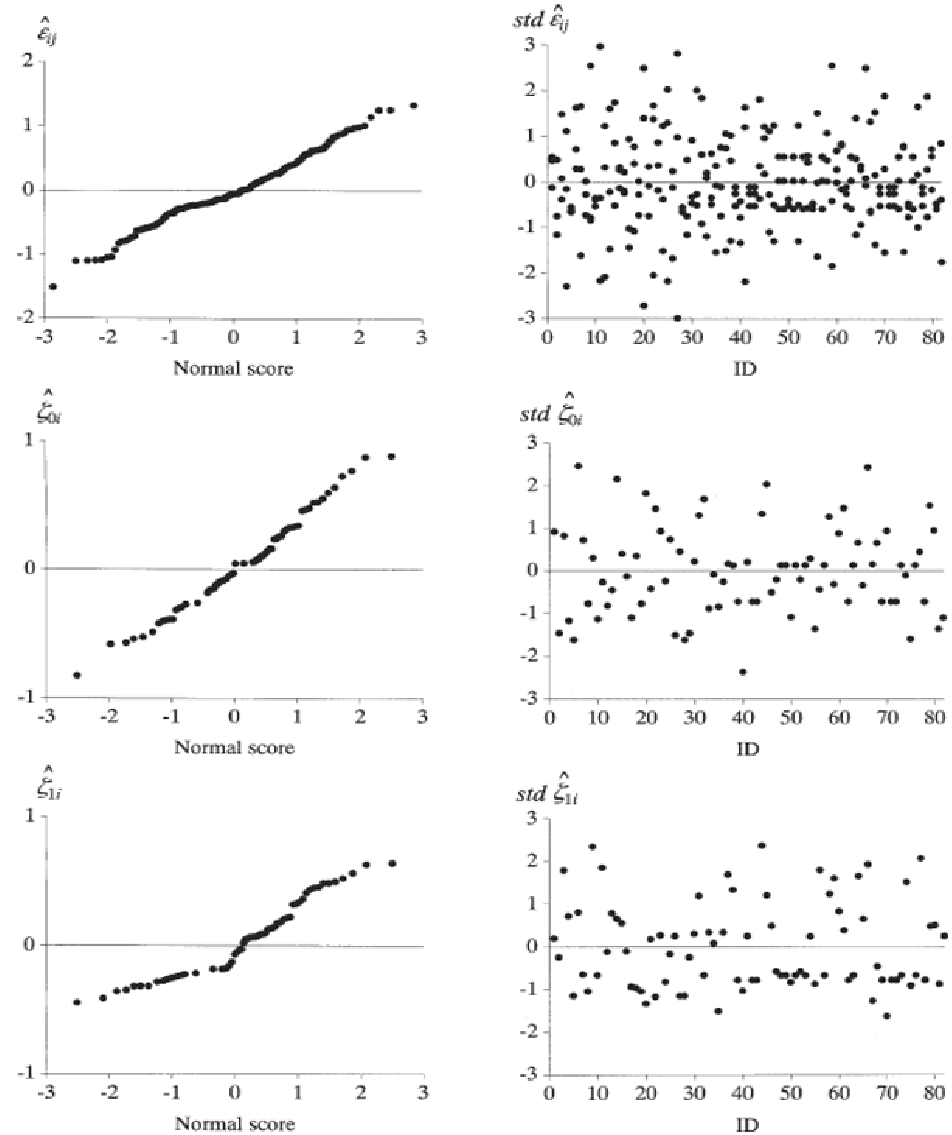


# Checking Normality Assumption

(Singer and Willett, pg. 130)

- Want to make sure the residuals are normally distributed
- Any departure from the line is a threat to normality.
- The std of the residuals should be scatter with no form around 0 and 95% of scatter within 2 stdv of 0
- It looks like, for **MODEL F,** the level-1 residuals are normal
- Level-2 residual may be concerning



# Checking Normality Assumption

(Singer and Willett, pg. 130 [STATA version – see UCLA site])

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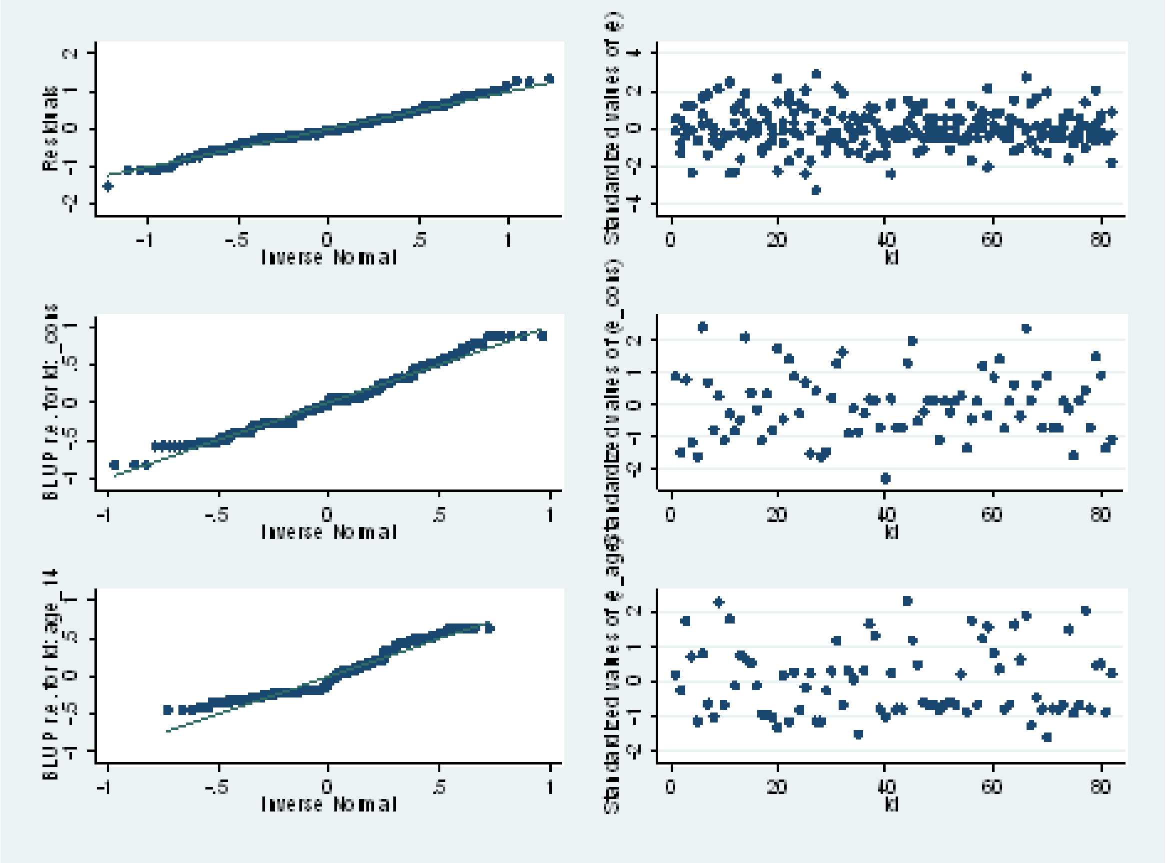
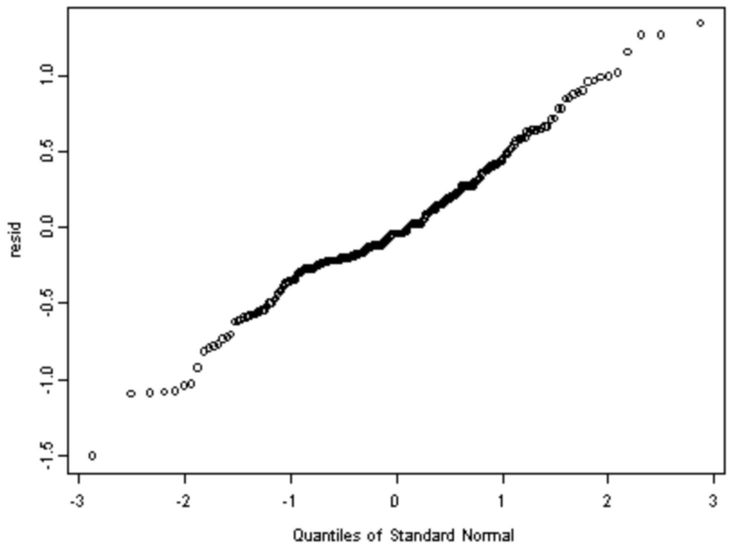



https://stats.idre.ucla.edu/stata/examples/alda/chapter3/applied-longitudinal-data-analysis-modeling-change-and-event-occurrenceby-judith-d-singer-and-john-b-willett-chapter-3-introducing-the-multilevel-model-for/

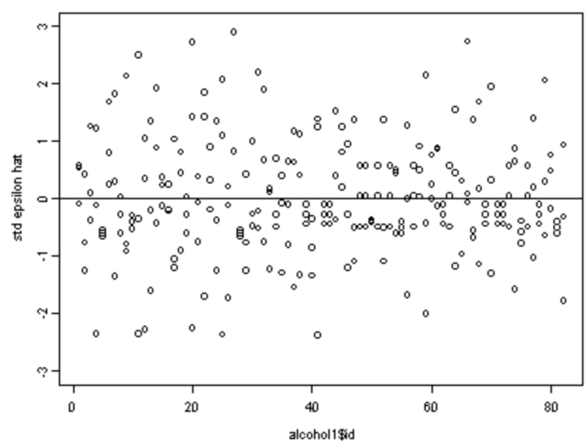
# Checking Normality Assumption

(Singer and Willett, pg. 130 [R version – see UCLA site])

```
#creating the residuals (epsilon.hat)
resid <- residuals(model.f)
qqnorm(resid)
```




```
#creating the standardized residual (std epsilon.hat)
resid.std <- resid/sd(resid)
plot(alcohol1$id, resid.std, ylim=c(-3, 3), ylab="std epsilon hat")
abline(h=0)
```

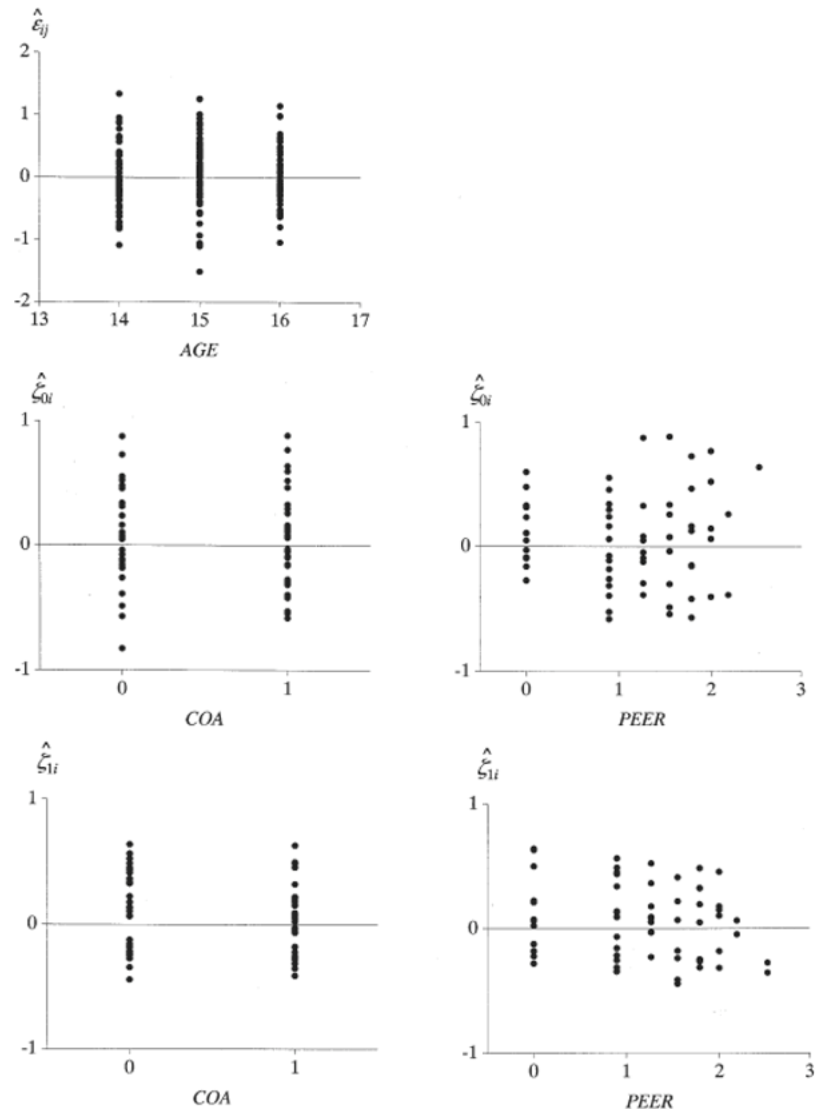



https://stats.oarc.ucla.edu/r/examples/alda/r-applied-longitudinal-data-analysis-ch-4/

# Checking Homoscedasticity

(Singer and Willett, pg. 133)

- Plotting the level-1 residuals against level-1 predictors
- Plotting the level-2 residuals against level-2 predictors
- If assumption holds the residual variance will be equal at each level of the predictor
- Level-1 and level-2 residuals seems to have equal range and variability at each time point
- At higher levels of PEERS there seems to be a drop in variability with predictor values greater than 2.5. This may suggest heterogeneity



# Checking Homoscedasticity

(Singer and Willett, pg. 130 [STATA version – see UCLA site])

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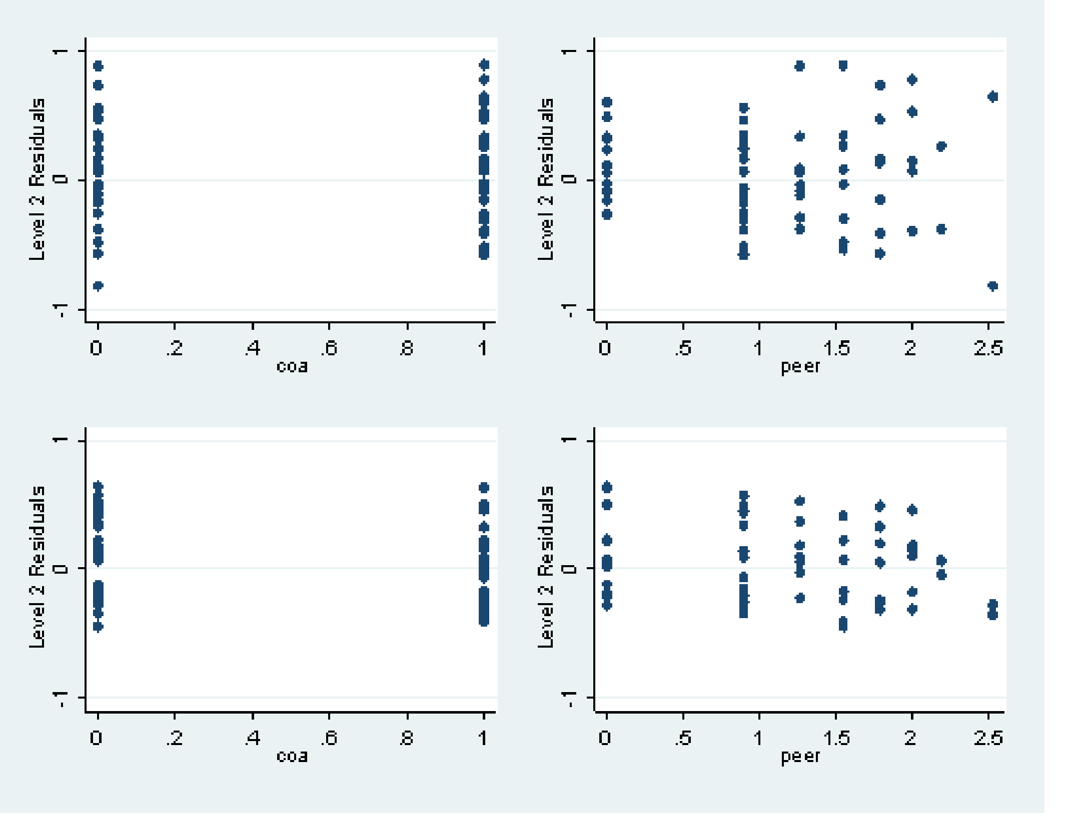


https://stats.idre.ucla.edu/stata/examples/alda/chapter3/applied-longitudinal-data-analysis-modeling-change-and-event-occurrenceby-judith-d-singer-and-john-b-willett-chapter-3-introducing-the-multilevel-model-for/

# Summary of Testing Assumptions

- Check for functional form, normality and homoscedasticity
- Our estimators can handle a bit of deviance in our assumptions, but you must do additional post hoc testing to make sure that your are receiving unbiased estimates
- When these assumptions are violated we may have to find alternative approaches
  - E.g. Non-linear model

# Administrative Stuff

# Final Project

- We will ask you to sign up as teams for final course projects

- Each team will submit a 1-2 page proposal before the spring break
  (ONE Proposal Per Team. All teams will receive full credits on the proposal as long as you submitted the required informations)

# Next Class…

- Estimators
- Time-varying covariates
  - **Singer and Willet pgs. 159- 181**

- Discuss assignment #3

# Class Exercises

# Class Exercise 1: Assessing Model Fit and Testing Assumptions

1. Run the unconditional mean model, the unconditional growth model, and the “final” model with the covariate biological sex (may have to recode so it makes sense)
2. Determine the best model fit based on what you learned in todays lecture
3. Test the model assumptions for the “final” model

https://stats.idre.ucla.edu/stata/examples/alda/chapter4/applied-longitudinal-data-analysis-modeling-change-and-event-occurrenceby-judith-d-singer-and-john-b-willett-chapter-4-doing-data-analysis-with-the-multile/

# Class Exercise 2: Assessing Model Fit and Testing Assumptions

- Flint Adolescent Study
- Mean depression at year 1-5 (15-19 years of age)
- Sex (male; female)
- Directions same as ”Class Exercise 1”